DIA 2025 - Competition for Development Measuring Market Power in LAC

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Abstract

This document offers a comprehensive explanation of the definitions and calculations for all indicators used in the Development in the Americas 2025 report on Competition for Development. Its objective is to evaluate market competition in Latin America and the Caribbean, focusing on both its level and evolution. The included metrics enable a thorough analysis of competition and market power across the region, ensuring consistency and comparability. These indicators are categorized into four modules: Concentration, Firm Dynamics, Market Power, and Firm Performance.

1 Introduction

The analysis of market structure and competition requires a rigorous and multidimensional understanding of several factors all at the same time. To implement a complete analysis we need to take advantage of some of the most used indicators available in the literature. For example, the Herfindahl-Hirschman Index (HHI) is a commonly used indicator that reflects market concentration but it doesn't account for entry and exit. At the same time, markups and markdowns are excellent measures to

capture market power, but it doesn't explain if the level of market power observed is "good" or not.

Hence, to perform a complete diagnosis of markets we will take advantage of different aspects involved in market competition and its structure. We identify four key aspects or categories involved in the analysis of markets. Concentration, Firm Dynamics, Market Power, and Performance. A joint analysis where each category provides a specific lens to examine different aspects of market competition proves to give a more complete interpretation of the results.

1. **Concentration**: We assess the market structure, focusing on the level of dominance held by individual firms or small groups of players at the sector level[1]. This provides insights into the potential for firms to exert undue influence over prices and market outcomes.

2. **Firm Dynamics**: Dynamism offers a valuable perspective on the continuous evolution of markets over time. By assessing key market measures, including the entry and exit rates of firms, the market share reshuffling, and turnover at the top, we gain insights into the dynamic nature of markets. These indicators provide a comprehensive picture of market dynamism, revealing the fluctuations and transformations in market structure over time. It also allows for the identification of shocks at both the national and market levels.

3. **Firm Performance**: We identify firm perfomance using productivity (TFP), labor misallocation, and profitability of firms. These indicators provide insights into the overall performance of firms, and identifying potential areas for improvement.

4. **Market Power**: We delve into the ability of firms to influence prices and quantities beyond what would be expected in a perfectly competitive market, directly impacting consumer choices and welfare. To achieve this, we calculate indicators at the firm level to assess their influence in the market they operate. Some of the included indicators are markups, price-cost margin, and labor markdown.

[1]We explore various definitions of sector-level such as: product-level through Economic Industry Classification, and geographic-level via country-specific disaggregation or any relevant market definition for our study.

2 Hardware and Software

Programs were implemented on a Windows operating system using STATA 14. Users may also need to install supplemental STATA packages. If you get any errors, please check if it is needed to install a package in the `_Begin.do`. All the packages required are also available in the 'ado' folder.

3 Code and Replication Instructions

The replication materials are comprised of several STATA do-files for each module. The program `_MASTER.do` executes all these do-files and reproduces the results. Tables are exported in .dta, .csv or .xlsx format and figures are in .png format.

Users are required to customize `_BEGIN.do` to incorporate country-specific characteristics of the dataset. These include the country's name, initial and final year, the finest level of disaggregation within sectors, and firm and plant IDs (if available). Furthermore, users must rename essential variables such as year, sector, sales, labor, gross output, wage bill, capital, and materials, along with any additional variables such as exports, imports, and geographic location, which will be utilized to compute the indicators in the code.

4 Output

Results are exported in the folders 'figures' and 'tables' inside the 'display' folder.

5 Market Competition Indicators

5.1 Module 1 - Concentration Measures Indicators

Concentration measure is an important part of assessing how good or bad a market is performing. Nevertheless, there is no consensus on how to perfectly measure concentration. There exists a variety of indicators that shed light on how to measure concentration in markets. For example, Herfindahl-Hirschman Index (HHI) is a

well-known and used indicator that identifies how concentrated is a market based on the revenue market shares. As the HHI we have several other indicators such as Concentration Revenues (CR) of top 4 firms, commonly known as CR4. Revenue Market Share, Market Share of Top 1%, Share of Bottom 50%, and Dominance Index are some of the indicators we pretend to use for this study. We present a list of the indicators used for this analysis and their respective formula.

We define the following notation to be considered in all the formulas used in this document. We index:

- Firm with $i \in N$.
- Sector with $j \in J$.
- Year with $t \in T$.

1. **Revenue Market Share**: A company's market share represents its relative size in an industry. We determine a company's market share by dividing its total sales or revenues by the industry's total sales in a fiscal period. We are interested in the mean, median, and standard deviation of this measure. To compute, we have the following equation for firm i in sector j in period t:

$$\text{Market Share}_{ijt} = \frac{\text{revenue}_{ijt}}{\sum\limits_{i \in j} \text{revenue}_{ijt}} \tag{1}$$

2. **Revenue Market Share (adjusted for trade component, if available)**: is a measure that provides a more detailed and accurate view of a company's market position, adjusting it for specific factors related to its trade activities. Essentially, the adjustment in the market share is made by subtracting the firm's exports in the numerator and the sector's net exports (exports minus imports) in the denominator. This means that it is considering domestic market revenue as a percentage of the sector's total revenue adjusted for trade. This allows us to better understand a company's competitive position within a local market, adjusting for the impact of international trade. The adjusted Market Share^{adj} for firm i in sector j at period t can be calculated using the following formula:

$$\text{Market Share}^{\text{adj}}_{ijt} = \frac{\text{revenue}_{ijt} - \text{export}_{ijt}}{\sum\limits_{i \in j} \left[\text{revenue}_{ijt} + \left(\text{import}_{ijt} - \text{export}_{ijt}\right)\right]}, \tag{2}$$

Notice that we can rewrite the Market Share$^{\text{adj}}_{ijt}$ abstracting the exports from the numerator (export$_{ijt}$) as:

$$\text{s}^{\text{adj}}_{ijt} = \frac{\text{sales}_{ijt}}{\left[\text{sales}_{jt} + \text{import}_{jt} - \text{export}_{jt}\right]} = \alpha_{jt} \times \text{sales}_{ijt} \tag{3}$$

where $\alpha_{jt} = \frac{1}{\text{sales}_{jt}+\text{import}_{jt}-\text{export}_{jt}}$ is firm-level invariant. Hence we can separate the market share into two components. The first component is the firm-level variant (s_{ijt}) and the second component is the sector-year variant (α_{jt}). Hence, the adjusted market share can be expressed as:

$$\text{s}^{\text{adj}}_{ijt} = \alpha_{jt} \times \text{sales}_{ijt} = \underbrace{\alpha_{jt} \times \text{sales}_{jt} \times \frac{\text{sales}_{ijt}}{\text{sales}_{jt}}}_{\substack{\text{multiplying by sales}_{jt} \\ \text{in the numerator and denominator}}} = \alpha_{jt} \times \text{sales}_{jt} \times \text{s}_{ijt} = \hat{\alpha}_{jt} \times \underbrace{\text{s}_{ijt}}_{\substack{\text{defined in} \\ \text{equation (1)}}} \tag{4}$$

where $\text{s}_{ijt} = \frac{\text{sales}_{ijt}}{\text{sales}_{jt}}$ and $\hat{\alpha}_{jt} = \frac{\text{sales}_{jt}}{\text{sales}_{jt}+\text{import}_{jt}-\text{export}_{jt}}$.
So the HHI can be written as:

$$HHI^{\text{adj}}_{jt} = \sum_{i \in j} [\text{s}^{\text{adj}}_{ijt}]^2 = \sum_{i \in j} \hat{\alpha}^2_{jt} \times \text{s}^2{}_{ijt} = \hat{\alpha}^2_{jt} \sum_{i \in j} \text{s}^2{}_{ijt} = \hat{\alpha}^2_{jt} \times HHI_{jt} \tag{5}$$

3. **Herfindahl-Hirschman Index (HHI)**: is a commonly used measure to identify market concentration. It is calculated by summing the square of the market shares of all firms within a sector. The higher the HHI, the more concentrated the market is. Here is the formula for the HHI for sector j in period t:

$$\text{HHI}_{jt} = \sum_{i \in j} (\text{Market Share}_{ijt})^2 \tag{6}$$

We can also compute the HHI accounting for the trade component by using the adjusted Market Share indicator (equation 5).

4. **Herfindahl-Hirschman Index (decomposition)**: the HHI can be decomposed into two factors: i) the variance of the size distribution of firms, and ii) the number of operating firms. Intuitively, when the number of firms is constant, the variance of the firm size distribution determines the magnitude of market concentration. Likewise, if all firms have a similar size, the presence of a hundred firms means a far less concentrated

environment than a market with only a few enterprises. Thus, we can think of HHI as the sum of the actual market state's deviation from 1) all firms having the same size, and the deviation from 2) a fully competitive environment with an infinite number of firms in the market. The decomposition of the HHI for sector j in period t can be written as:

$$\text{HHI}_{jt} = E^i_{jt} + E^n_{jt} \tag{7}$$

where $E^i = \tau^2$ and $E^n = (\text{HHI} - \tau^2)$. Moreover, we can define $\tau^2 = (n_j \times \text{HHI} - 1)/(n-1)$. Thus, this decomposition produces two components explaining the observed market concentration, E^i_{jt} , the inequality of firm sizes effect, and E^n_{jt} , the number of firms effect.

5. **Concentration Ratio (CR)**: is another commonly used metric in economics to assess the degree of concentration within an industry. It is typically expressed as the total percentage of market share held by the largest firms within the industry. The most common concentration ratios are the CR4 and CR8, which represent the market share of the top 4 and top 8 largest firms, respectively. Concentration ratios provide a glance at the market structure and competitive landscape. A high concentration ratio indicates that a few companies dominate the market, which could suggest an oligopolistic market. In contrast, a low concentration ratio implies more competition among a larger number of small firms.

$$\text{CR}^K_{jt} = \sum_{k=1}^{K} \text{Market Share}^{\#k}_{jt} \quad \text{for K} = \{4, ..., 10\} \tag{8}$$

where Market Share$^{\#k}_j$ represents the $k$est highest market share in sector j in period t.

6. **Share of top 1%, 10%, and 25%**: this indicator measures market concentration similarly to concentration ratios but is based on the firm's percentile. It measures the share of the top 1%, 10%, or 25% of top firms in a certain market. A higher measure of this indicator suggests that the top 1%, 10%, or 25% control a significant portion of the market share, and the market might be less competitive and more susceptible to monopolistic or oligopolistic behavior. Conversely, a lower measure of this indicator suggests a more competitive market environment with less concentration.

$$\text{Share Top K\%}_{jt} = \sum_{i \in \text{Top K\%}} \text{Market Share}_{ijt}, \quad \text{K} = \{1, 10, 25\} \tag{9}$$

7. **Share of bottom 50%**: this indicator refers to the collective market share of the bottom 50% of firms in the market. This measure highlights the extent to which smaller players, which constitute half of the number of firms, can compete in the market. A small collective share might indicate that the market is highly concentrated among the top firms, while a larger share would suggest a more competitive environment with numerous firms able to maintain significant market presence. One way to compute it is the following:

$$\text{Share Bottom 50\%}_{jt} = \sum_{i \in \text{Bottom 50\%}} \text{Market Share}_{ijt} \tag{10}$$

for sector j in period t.

8. **Dominance Index**: is another measure of market concentration, similar to the HHI, but with a focus on the disparity between firms in terms of their market share. It is used to gauge the dominance of companies within a market, considering not just the size of firms but also the gaps between them. The Dominance Index sums the squares of the differences in market shares between ranked firms in a sector.[2] A higher value indicates that a small number of firms have much larger market shares compared to the rest, implying a more dominant position in the market. Here is how the Dominance Index (DI) is calculated for sector j at period t:

$$\text{DI}_{jt} = \sum_{i=1}^{N-1} (s_i - s_{i+1})^2 \tag{11}$$

where s_i is ordered market share of firm i from largest to smallest (i.e. $s_1 \geq s_2 \geq \ldots \geq s_i \geq s_{i+1} \geq \ldots \geq s_N$) within sector j at period t.

5.2 Module 2 - Firm Dynamism Indicators

Dynamism Indicators are metrics used to assess the evolution of competition across time in a market or industry. These indicators help understand how dynamic and competitive a market is by tracking the movement and behavior of firms within it. A market characterized by high turnover at the top, high market share reshuffling, and significant entry and exit rates is typically considered dynamic and competitive. Such markets are

[2] Firms within a sector are ordered or ranked in descending order according to their revenue-based market share.

often associated with rapid innovation, improved customer services, and changing market leaders. Next we present some of these indicators and how to compute them:

1. **Turnover at the Top**: How likely is that one of the top firms (top 4) will exit the top ranking over the next k years? This measures the proportion of firms that have left the top 4 ranking relative to the total number of firms that were in the top 4 at the starting period. The term "exiter" refers to a firm that was in the top 4 at the beginning of the period but is no longer in the top 4 at the end of the period (notice that the firm could still be producing but no longer part of the top 4 firms in the market). For instance, if you start with 4 firms at the top in period t and 1 of those firms is no longer in the top 4 by period $t+k$, the Turnover at the Top would be 25%. The formula for calculating "Turnover at the Top" for a sector j in period t is as follows[3]:

$$\text{Turnover at the top}_{jt} = \frac{\text{\# of exiters}_j \text{ in top 4 from } t \text{ to } t+k}{\text{\# of firms}_j \text{ in top 4 from } t} \tag{12}$$

$$\text{Turnover at the top}_{jt} = \Pr\left(\text{Mkt Sh.}_{i,j,t+k} < \text{Mkt Sh.}^{\#4}_{j,t+k} | \text{Mkt Sh.}_{i,j,t} \geq \text{Mkt Sh.}^{\#4}_{j,t}\right) \tag{13}$$

where $\text{Mkt Sh.}^{\#4}_{j,t}$ represents the fourth highest market share in sector j in period t. A high turnover could indicate a competitive and dynamic sector where firm innovation can change significantly, while a low turnover could indicate a more stable, perhaps less competitive market structure where the top firms maintain their positions consistently.

2. **Market Share Reshuffling (of incumbents)**: refers to the changes in the ranking of firms by revenue within a market over time. The idea is to calculate a reshuffling index based on the correlation of firms' ranks at two different points in time. The reshuffling index is thus a measure of market dynamism, with higher values indicating a more dynamic and potentially competitive market where firms frequently change their standing, and lower values suggesting a more static market structure. For sector j in period t, we have:

$$\text{Reshuffling}_{jt} = 1 - \text{corr}(\text{Rank}_{j,t}, \text{Rank}_{j,t+k}) \quad \text{for } k = \{3, 5, 8\} \tag{14}$$

[3]We will impose some restrictions in this indicator to allow for comparisons between sectors and countries. The first restriction is that we will only take four top firms in each sector. In case we have ties in the top 4 firms resulting in more than 4 firms we will discard that sector from our analysis. The second restriction will be on sectors with less than 4 firms (hence we have less than 4 firms in the top 4). We will not consider sectors with less than 4 firms in total. Nevertheless, we consider that the cases where these restrictions will be applied are going to be very low.

The formula implies that as the correlation decreases (indicating more reshuffling), the reshuffling index increases. The index is designed to increase as the stability of firms' ranks decreases. The index k represents the time lag between the two periods being compared. In this case, the lags are 3, 5, and 8 periods (years). This allows for the analysis of short-term versus long-term stability in market rankings.

3. **Entry Rate**: is a metric used to gauge the level of new business activity within a particular sector over a given time period. It is an indicator of the dynamism and competitiveness of the market. A high entry rate may suggest that the market is attractive, with low barriers to entry or high potential profits, while a low entry rate might indicate a saturated market or high barriers to entry. To calculate the entry rate:

$$\text{Entry Rate}_{jt} = \frac{\text{\# entrants}_{jt}}{\text{\# producing firms in } j,t}, \quad \forall j,t \tag{15}$$

where $\#\text{entrants}_{jt}$ represents the number of new firms that have started producing in sector j during period t. These are firms that were not producing in the previous period $(t-1)$. Moreover, $\#$producing firms in j,t is the total number of firms that are active and producing in sector j during the period t, including both new entrants and incumbent firms. For instance, if 10 new firms started producing in sector j in 2024 (period t) and there are 100 firms in total producing in sector j in 2024, then the entry rate would be 10%.

4. **Exit Rate**: is a concept related to the entry rate presented above. The exit rate measures the rate at which firms leave a market or sector. A higher exit rate suggests a larger proportion of firms leaving the market, which might be a sign of a highly competitive or declining sector. Conversely, a lower exit rate indicates relative market stability with fewer firms leaving. An "exiter" is a firm that was producing goods or services in one time period (say, year t), but has stopped producing in the subsequent period (year $t+1$). This cessation of production could be due to various reasons such as bankruptcy, mergers, shifts in business strategy, or other economic factors that made it unsustainable for the firm to continue operating in that sector. To calculate the exit rate:

$$\text{Exit Rate}_{jt} = \frac{\text{\# exiters}_{jt}}{\text{\# producing firms in } j,t}, \qquad \forall j,t \tag{16}$$

5. **Establishment size**:[4] this measure is calculated at the establishment/firm level. We take into account the panel component in our dataset and calculate a simple average of employment in t and $t-1$. We define this metric at establishment i in period t as:

$$x_{it} = \frac{\text{employment}_{it} + \text{employment}_{it-1}}{2} \tag{17}$$

Notice that this measure can be obtained at the sector level j analogously. We can also aggregate this measure to year-level using the following formula:

$$\text{X}_t = \sum_{j \in J} X_{jt} = \sum_{j \in J} \sum_{i \in j} x_{it} \tag{18}$$

6. **Employment Growth Rate**: this indicator shows the employment growth at the establishment/firm level. This measure oscillates between $[-2, +2]$, and it identifies death (when the indicator is equal to -2) and birth (when the indicator is equal to $+2$) of the firm. We define this indicator at the firm i in period t as:

$$\text{g}_{it} = \frac{\text{employment}_{it} - \text{employment}_{it-1}}{x_{it}} \tag{19}$$

Notice that this metric is related to the traditional growth rate ($G \equiv \frac{2g}{2-g}$). We can also aggregate this measure to the year-level using the following formula:

$$\text{g}_t = \sum_{j \in J} \left(\frac{X_{jt}}{X_t} \right) g_{jt} = \sum_{j \in J} \left(\left(\frac{X_{jt}}{X_t} \right) \sum_{i \in j} \left(\frac{x_{it}}{X_{jt}} \right) g_{it} \right) \tag{20}$$

7. **(Gross) Job Creation Rate**: it identifies how many firms have had positive growth rates as a measure of job creation in a sector j at period t. To calculate it, we have to account for the total employment (X_{jt}) in sector j at t, the firm level size (x_{it}), and firms with positive employment growth (g_{it}). Consider the job creation rate at sector j at t as:

$$\text{POS}_{jt} = \sum_{\substack{i \in j_t \\ g_{it} > 0}} \left(\frac{x_{it}}{X_{jt}} \right) g_{it} \tag{21}$$

We can also decompose the POS_{jt} indicator. The job creation from entry at the firm level

[4] This indicator and the following presented below for firm dynamism are taken from Haltiwanger's paper (Gross Job Creation, Gross Job Destruction, and Employment Reallocation, 1992 QJE)

is given by:

$$\text{POS_Entry}_{jt} = \sum_{\substack{i \in j_t \\ g_{it} > 0}} \left(\frac{x_{it}}{X_{jt}} \right) g_{it} \times I\{g_{ijt} = 2\} \tag{22}$$

where I is an indicator function equal to one if expression in brackets hold, zero otherwise, and $g_{ijt} = 2$ denotes an entrant. Moreover, we can also compute job creation only for continuing firms:

$$\text{POS_Cont}_{jt} = \sum_{\substack{i \in j_t \\ g_{it} > 0}} \left(\frac{x_{it}}{X_{jt}} \right) g_{it} \times I\{g_{ijt} \in (0, 2)\} \tag{23}$$

Given these definitions, the following simple relationships hold:

$$\text{POS}_{jt} = \text{POS_Entry}_{jt} + \text{POS_Cont}_{jt} \tag{24}$$

8. **(Gross) Job Destruction Rate**: analogous we can calculate the job destruction rate. It identifies how many firms have had negative growth rates as a measure of job destruction in a sector j at period t. To calculate it, we have to account for the total employment (X_{jt}) in sector j at t, the firm level size (x_{it}), and firms with negative employment growth (g_{it}). Consider the job destruction rate at sector j at t as:

$$\text{NEG}_{jt} = \sum_{\substack{i \in j_t \\ g_{it} < 0}} \left(\frac{x_{it}}{X_{jt}} \right) |g_{it}| \tag{25}$$

Again, we can decompose the NEG_{jt} indicator. The job destruction from exit at the firm level is given by:

$$\text{NEG_Exit}_{jt} = \sum_{\substack{i \in j_t \\ g_{it} < 0}} \left(\frac{x_{it}}{X_{jt}} \right) g_{it} \times I\{g_{ijt} = -2\} \tag{26}$$

where I is an indicator function equal to one if expression in brackets hold, zero otherwise, and $g_{ijt} = -2$ denotes an exit. Moreover, we can also compute job destruction

only for continuing firms:

$$\text{NEG_Cont}_{jt} = \sum_{\substack{i \in j_t \\ g_{it} < 0}} \left(\frac{x_{it}}{X_{jt}} \right) g_{it} \times I\{g_{ijt} \in (-2, 0)\} \tag{27}$$

Given these definitions, the following simple relationships hold:

$$\text{NEG}_{jt} = \text{NEG_Exit}_{jt} + \text{NEG_Cont}_{jt} \tag{28}$$

Notice that the job creation and job destruction rates directly reflect the reallocation of employment positions (or jobs) and not the reallocation of workers. We can think of these as a reallocation of employment measures since year-to-year changes in a firm's total employment are mainly induced by optimal firm size rather than temporary movements in the stock of unfilled positions.

9. **(Gross) Job Reallocation Rate**: identifies the gross job reallocation rate in sector j at between $t-1$ and t. Notice that the job creation and the job destruction rates reflect reallocation measures (either creation or destruction) of jobs in a certain sector. Hence, a natural indicator of the overall job reallocation in sector j can be obtained as:

$$\text{SUM}_{jt} = \text{POS}_{jt} + \text{NEG}_{jt} \tag{29}$$

We can also decompose the SUM_{jt} in the following:

$$\text{SUM}_{jt} = (\text{POS_Entry}_{jt} + \text{POS_Cont}_{jt}) + (\text{NEG_Exit}_{jt} + \text{NEG_Cont}_{jt}) \tag{30}$$

$$= (\text{POS_Cont}_{jt} + \text{NEG_Cont}_{jt}) + (\text{POS_Entry}_{jt} + \text{NEG_Exit}_{jt}) \tag{31}$$

$$= \text{SUM_Cont}_{jt} + \text{SUM_Entry}_{jt} + \text{SUM_Exit}_{jt} \tag{32}$$

In terms of the frequency of distribution, the reallocation rate can be thought of as the size-weighted mean of the absolute value of firm growth rates. When multiplied by the sector size, i.e.,

$$X_{jt}\text{SUM}_{jt} = X_{jt}(\text{POS}_{jt} + \text{NEG}_{jt}) \tag{33}$$

it represents an *upper bound* on the number of workers who change jobs or switch between

employment and non-employment in response to firm-level employment changes.

To obtain a *lower bound* on the number of workers who switch between jobs or job status, we have to eliminate the possibility of double counting job losers who move directly to new jobs at expanding establishments in the same sector. That is,

$$\mathrm{X}_{jt}\mathrm{MAX}_{jt} = \mathrm{X}_{jt} \max\{\mathrm{POS}_{jt}, \mathrm{NEG}_{jt}\} \tag{34}$$

represents a lower bound on the number of workers who change jobs or employment status in direct response to job reallocation in sector j.

10. **Net Employment Growth Rate**: is the difference between the job creation rate and the job destruction rate. It shows the net job balance in sector j at t. This measure could be positive if job creation was bigger than job destruction (hence a positive growth or expansion in employment), or negative if job destruction was bigger than job creation (hence a negative growth or shrink in employment). The Net Employment Growth for sector j is calculated as:

$$\mathrm{NET}_{jt} = \mathrm{POS}_{jt} - \mathrm{NEG}_{jt} \tag{35}$$

Notice that this measure can be considered a *lower bound* for net employment growth since we cannot determine whether a given level of employment in two different periods for the same firm represents the same or different employment positions.

11. **Excess Job Reallocation Rate**: is the mean difference between total job reallocation and the minimum job reallocation required to accommodate net employment growth changes. It is calculated as the difference between SUM_{jt} and the absolute value of NET_{jt}.

$$\mathrm{EXC}_{jt} = \mathrm{SUM}_{jt} - \mid \mathrm{NET}_{jt} \mid \tag{36}$$

12. **Fraction of persistence of job creation/destruction**: is an indicator of job persistence in the economy. It measures the fraction of jobs created between March of year $t-1$ and March of year t that persists through March of year $t+k$.[5] [6] The idea of this indicator

[5]Haltiwanger stresses the importance of measuring the beginning and end of this indicator indexed by the same month. The idea is to control for temporal shocks associated with labor dynamics and for more accuracy of this measure.

[6]It counts only the jobs created in t that persist in $t+k$.

is to account for the character of worker reallocation associated with job reallocation. Since job destruction and creation can be perceived as a short-term firm-level employment change (year-to-year basis), we need to determine how much of this reallocation has a long-term persistence at the firm-level. These persistence changes are associated with long-term joblessness or worker reallocation across plants. We can calculate the fraction of job persistence of job creation (FPOS) and destruction (FNEG) in t that continue to be present in $t+1$ as:

$$\text{New Jobs}_{it} = \text{employment}_{it} - \text{employment}_{it-1} \tag{37}$$

$$\text{FPOS}_{t1} = \sum_{\substack{i \in J_t \\ \text{New Jobs}_{it} > 0}} \frac{\text{New Jobs}_{it}}{\text{POS}_t} \tag{38}$$

$$\text{FNEG}_{t1} = \sum_{\substack{i \in J_t \\ \text{New Jobs}_{it} < 0}} \frac{\text{New Jobs}_{it}}{\text{NEG}_t} \tag{39}$$

Notice this indicator can be calculated for $t+2$ following the same logic presented before. Thus, FPOS_{t2} denotes the fraction of newly created jobs in t that are present in $t+1$ *and* $t+2$. FNEG_{t2} is defined analogously.

13. **Decomposition of excess job reallocation**: accounts for disturbances in the employment reshuffling caused by sectors and firms simultaneously creating and destroying jobs. Since we could have a firm in a period t hiring new workers and laying off old workers at the same time, the effect on the total job reallocation could be biased since it doesn't account for these scenarios. We can decompose the job reallocation into two components. One component captures the *between* sector effect, and the other component represents the contribution of excess job reallocation *within* sectors. The following are the formulas for the *between* and *within* components.

The component of excess job reallocation due to *between-sector* employment shifts is given by

$$\sum_{j=1}^{J} \Big(\mid \text{Net Employment Change in } j \mid \Big) - \mid \underbrace{\text{Overall Net Employment Change}}_{\sum_{j=1}^{J} \text{Net Employment Change in } j} \mid \tag{40}$$

In the other hand, the component due to excess job reallocation *within-sectors* is given by

$$\sum_{j=1}^{J}\Big(\text{Job Reallocation in } j- \mid \text{Net Employment Change in } j \mid\Big) \tag{41}$$

5.3 Module 3 - Performance Indicators

1. **Productivity (TFP)**: extensively described in the next section.

2. **Value Added per Worker**: is a classical measure of labor productivity since it divides the value added generated by the firm among the employees used to generate total output. It is calculated at the firm-level:

$$\text{Value Added per Worker}_{ijt} = \frac{\text{Value Added}_{ijt}}{\text{Employment}_{ijt}} \tag{42}$$

3. **Firm Profit:** refers to the financial gains a company makes after accounting for all its expenses. It is a crucial indicator of a company's financial health and its efficiency in managing its operations and resources. We can compute the firm profit as follows:

$$\text{Firm Profit}_{ijt} = \text{Total Revenue}_{ijt} - \text{Total Costs}_{ijt} \tag{43}$$

for firm i in sector j in period t.

4. **Profit Rate:** refers to the ratio between the firm's profits and its total revenue. We can compute the profit rate as follows:

$$\text{Profit Rate}_{ijt} = \frac{\text{Firm Profit}_{ijt}}{\text{Total Revenue}_{ijt}} \tag{44}$$

for firm i in sector j in period t.

5.4 Module 4 - Market Power Indicators

There are two prominent methods for calculating markups, the demand approach and the production approach. The demand approach relies on a fully specified model of consumer choice to derive a demand system that can be estimated using data on market shares, prices, and product attributes. Given the estimated own and cross-price

elasticities across the goods considered, markups can be recovered from first-order conditions after specifying a model of competition (Berry et al., 1995; Berry et al., 2004). The production approach relies on data on input and output levels and requires that producers choose variable input levels to minimize costs.

When implementing the demand approach, researchers must specify the way in which firms in the market compete (e.g., static Nash Bertrand). The notion of market equilibrium, together with estimates of demand elasticities, allows one to infer marginal costs and, therefore, markups. To estimate demand elasticities, researchers typically specify a model of consumer heterogeneity that restricts the pattern of cross-price elasticities to some extent, and instrumental variables are needed for consistent estimation of demand elasticities, given the potential correlation between prices and demand shocks (De Loecker and Scott, 2016).

To implement the production approach, researchers must specify a variable input. Given estimated production function parameters, the conditions for cost-minimization with respect to the variable input allow researchers to compute the marginal cost of production and, therefore, markups. Estimating the production function requires an assumed functional form for the production technology and identifying assumptions to avoid bias from the potential correlation between productivity shocks and variable input levels.

5.4.1 Product Markups à la De Loecker et al. (2012, 2020)

Let's define the markup of firm i in period t as the ratio of the firm's good price (P_{it}) to its marginal cost (λ_{it}):

$$\mu_{it} = \frac{P_{it}}{\lambda_{it}}. \tag{45}$$

5.4.2 Markup estimation via Accounting Approach

Firm i at time t produces output using the following technology:

$$Q_{it} = Q_{it}(X_{it}^1, ..., X_{it}^J, K_{it}, \omega_{it}) \tag{46}$$

where $Q_{it}(.)$ is firm-specific technology, $X_{it} = (X_{it}^1, ..., X_{it}^V)$ is the vector of variable inputs of production (such as labor, intermediates, materials, electricity), K_{it} is the capital stock,

and ω_{it} is productivity.

The key assumption is that variable inputs frictionlessly adjust within one period (a year in most available databases), whereas capital is a dynamic input subject to adjustment costs and other frictions. The firm purchases inputs in perfectly competitive markets, taking factor prices as given. Instead, the firm has market power with respect to its own good, facing an arbitrary inverse demand schedule $P(Q)$.

Let's specialize in the case with a single variable input ($J = 1$) to ease exposition. The Lagrangian for cost minimization is given by:

$$\mathcal{L}_{it} = P_{it}^X X_{it} + r_{it}^K K_{it} + F_{it} - \lambda_{it}[Q_{it}(X_{it}, K_{it}) - \overline{Q}_{it}]$$

subject to (46), where P_{it}^X and r_{it}^K are the (potentially firm-specific) price of the variable input and the user cost of capital, F_{it} is the fixed cost, and the Lagrange multiplier λ_{it} is also a measure of the firm's marginal cost.

The first-order condition with respect to the variable input is given by:

$$\frac{\partial \mathcal{L}_{it}}{\partial X_{it}} = P_{it}^X - \lambda_{it} \frac{\partial Q_{it}}{\partial X_{it}} = 0$$

Multiplying both sides by $\frac{X_{it}}{Q_{it}}$:

$$\frac{P_{it}^X X_{it}}{Q_{it}} = \lambda_{it} \frac{\partial Q_{it}}{\partial X_{it}} \frac{X_{it}}{Q_{it}}$$

Using the markup definition $\mu_{it} = \frac{P_{it}}{\lambda_{it}}$ and rearranging, yields:

$$\mu_{it} = \frac{\left(\frac{\partial Q_{it}}{\partial X_{it}} \frac{X_{it}}{Q_{it}}\right)}{\left(\frac{P_{it}^X X_{it}}{P_{it} Q_{it}}\right)} \equiv \frac{\beta_{it}^X}{\alpha_{it}^X} \tag{47}$$

where β_{it}^X is the firm's output elasticity of the variable input, and α_{it}^X is the share of expenditures on the variable input in total revenues.

For instance, if the single variable input is labor ($X_{it} = L_{it}$, $P_{it}^X = W_{it}$), then:

$$\alpha_{it}^L = \frac{W_{it} L_{it}}{P_{it} Q_{it}} = \frac{\text{Wage Bill}_{it}}{\text{Total Revenue}_{it}}$$

Many databases such as Compustat, Refinitiv-Datastream, or Orbis, provide information for "Cost of Goods Sold" (COGS), which bundles expenses directly attributable to production, including materials, intermediate inputs, wage bill, electricity, and so on. For instance, De Loecker et al. (2020) (using Compustat) and Díez et al. (2021) (using Orbis) compute the revenue share of variable input as:

$$\alpha_{it}^{V} = \frac{\text{COGS}_{it}}{\text{Total Revenue}_{it}}$$

On the other hand, the output elasticities either β_{it}^{L} or β_{it}^{V} are not measured directly and have to be estimated using (production function) proxy methods addressing endogenous factors[7], or via cost shares[8].

5.4.3 Factor Elasticities via Cost Shares

The micro-foundations for using cost shares are as follows. Consider a gross output production function with two variable inputs, labor L_{it} and materials M_{it}:

$$Q_{it} = K_{it}^{\beta_{it}^{k}} L_{it}^{\beta_{it}^{l}} M_{it}^{\beta_{it}^{m}} \exp(\omega_{it}) \tag{48}$$

The profit maximization problem can be written as follows:

$$\Pi_{it} = \max_{Q_{it}} P_{it}(Q_{it})Q_{it} - C(Q_{it})$$

where $C(Q_{it})$ is the firm cost function defined by the cost minimization problem:

$$C(Q_{it}) = \min_{\{K_{it}, L_{it}, M_{it}\}} r_{it}^{K} K_{it} + W_{it} L_{it} + P_{it}^{M} M_{it} \quad \text{subject to (48).}$$

The first-order condition for profit maximization yields:

$$C'(Q_{it}) = P_{it} + \frac{\partial P_{it}}{\partial Q_{it}} Q_{it}$$

[7]See Olley and Pakes (1996), Levinsohn and Petrin (2003) and Wooldridge (2009).
[8]See Foster et al. (2016) and Edmond et al. (2018).

Dividing both sides by P_{it}, using the markup definition, and rearranging, yields:

$$P_{it} = \mu_{it} C'(Q_{it}) \quad \text{and} \quad \mu_{it} = \frac{1}{1+\varepsilon_{P,Q}} \quad \text{with} \quad \varepsilon_{P,Q} \equiv \frac{\partial P_{it}}{\partial Q_{it}} \frac{Q_{it}}{P_{it}}$$

Under monopolistic competition, firms charge a markup over marginal cost, and the firm's markup is a function of the elasticity of the inverse demand curve $\varepsilon_{P,Q}$: The lower the elasticity, the lower competition, and the larger the markup.

The Lagrangian for the cost minimization problem is given by:

$$\mathcal{L} = r_{it}^K K_{it} + W_{it} L_{it} + P_{it}^M M_{it} - \lambda_{it} [K_{it}^{\beta_{it}^k} L_{it}^{\beta_{it}^l} M_{it}^{\beta_{it}^m} \exp(\omega_{it}) - \overline{Q}_{it}]$$

Using $\lambda_{it} = C'(Q_{it}) = \frac{P_{it}}{\mu_{it}}$, the first-order conditions can be written as:

$$\begin{aligned} r_{it}^K K_{it} &= \beta_{it}^k \frac{P_{it} Q_{it}}{\mu_{it}} \\ W_{it} L_{it} &= \beta_{it}^l \frac{P_{it} Q_{it}}{\mu_{it}} \\ P_{it}^M M_{it} &= \beta_{it}^m \frac{P_{it} Q_{it}}{\mu_{it}} \end{aligned} \tag{49}$$

Defining returns to scale as $RTS_{it} = \beta_{it}^k + \beta_{it}^l + \beta_{it}^m$, the factor demand system (49) implies, for instance:

$$\beta_{it}^l = \frac{W_{it} L_{it}}{r_{it}^K K_{it} + W_{it} L_{it} + P_{it}^M M_{it}} RTS_{it} \tag{50}$$

Syverson (2011) argues against using firm-specific factor elasticities when factor adjustment costs are present. Further, for similar reasons, Foster et al. (2016) suggests using some time-averaging at the industry level. Thus, we impose two final simplifying assumptions to operationalize formula (50). Given data for N firms across T periods for any particular sector, (a) $\beta_{it}^l = \beta^l$ for all firms i and periods t in a given economic sector, with $\beta^l = T^{-1} \sum_{t=1}^T \beta_t^l$ and $\beta_t^l = N^{-1} \sum_{i=1}^N \beta_{it}^l$, and analogously, (b) $RTS_{it} = RTS$ for all firms in all sectors. An obvious benchmark is to assume $RTS = 1$ (CRS) and then do robustness using a couple of values $RTS < 1$ (DRS). Then provided data on the wage bill, material expenditures, and the stock of capital, we can recover output elasticities for variable inputs.

In principle, if no data is available, one could calibrate a common value β for all firms and

sectors. De Loecker et al. (2020) show following the latter strategy does not significantly change their results for the distribution and time evolution of markups across U.S. firms.

5.4.4 Factor Elasticities via Proxy Methods

As before, the goal is to obtain industry-level factor elasticities, this time estimating the production function. Consider again a production function of the form (48), imposing from the outset the assumption $\beta_{it}^j = \beta^j$, $j \in \{k, l, m\}$, for all i and t within a given sector.

Let $x = \log(X)$. Let y_{it} be the $\log$ of **observed** gross output, so that

$$y_{it} = q_{it} + \varepsilon_{it}$$

where ε_{it} is a zero-mean measurement error. Then, we can write (48) as follows:

$$y_{it} = \beta^0 + \beta^k k_{it} + \beta^l l_{it} + \beta^m m_{it} + \omega_{it} + \varepsilon_{it} \tag{51}$$

As is well-known, OLS estimation of (51) yields potentially biased estimates of the elasticities because of the likely correlation between input levels and (unobserved) firm-specific productivity. Intuitively, firms experiencing a large positive productivity shock may respond by using more inputs.

The timing assumptions to apply "Proxy Methods" are: (a) k_{it} and l_{it} are predetermined, chosen optimally in $t-1$, and (b) m_{it} is flexibly and optimally chosen in t, after firm observes ω_{it}. Then, the intermediate input's demand function is:

$$m_{it} = m(k_{it}, l_{it}, \omega_{it}).$$

Assuming monotonicity of the function $m(.)$, one can invert to write productivity as an (unknown) "control" function of observables:

$$\omega_{it} = m^{-1}(k_{it}, l_{it}, m_{it}).$$

Plugging the control function back into the production function:

$$\begin{aligned} y_{it} &= \beta^0 + \beta^k k_{it} + \beta^l l_{it} + \beta^m m_{it} + m^{-1}(k_{it}, l_{it}, m_{it}) + \varepsilon_{it} \\ &= \phi(k_{it}, l_{it}, m_{it}) + \varepsilon_{it} = \phi_{it} + \varepsilon_{it} \end{aligned}$$

with

$$\phi_{it} = \beta^0 + \beta^k k_{it} + \beta^l l_{it} + \beta^m m_{it} + m^{-1}(k_{it}, l_{it}, m_{it})$$

The first stage consists of approximating ϕ_{it} with a 2nd order polynomial, and run OLS on:

$$\begin{aligned} y_{it} = c &+ \beta_{1k} k_{it} + \beta_{1l} l_{it} + \beta_{1m} m_{it} + \beta_{2k} k_{it}^2 + \beta_{2l} l_{it}^2 + \beta_{2m} m_{it}^2 \\ &+ \beta_{kl} k_{it} l_{it} + \beta_{km} k_{it} m_{it} + \beta_{lm} l_{it} m_{it} + \varepsilon_{it} \end{aligned}$$

Removing the measurement error, projected output $\hat{\phi}_{it}$, is given by

$$\hat{\phi}_{it} = y_{it} - \hat{\varepsilon}_{it}$$

Then, for any $\beta = (\beta^0, \beta^k, \beta^l, \beta^m)$, we can write ω_{it} as a function of observables:

$$\begin{aligned} \omega_{it} &= m^{-1}(k_{it}, l_{it}, m_{it}) \\ &= \hat{\phi}_{it} - \beta^0 - \beta^k k_{it} - \beta^l l_{it} - \beta^m m_{it} = \omega_{it}(\beta) \end{aligned}$$

Assuming that firm-level productivity follows a 1st order Markov process

$$\omega_{it} = g(\omega_{it-1}) + \eta_{it}$$

In particular, one can approximate $g(.)$, for instance, using a 3rd order polynomial, and run OLS on:

$$\omega_{it} = \rho_0 + \rho_1 \omega_{it-1} + \rho_2 \omega_{it-1}^2 + \rho_3 \omega_{it-1}^3 + \eta_{it}$$

to obtain an estimate of $\hat{\eta}_{it}$.

Finally, the moment conditions to identify $\beta = (\beta^0, \beta^k, \beta^l, \beta^m)$ is given by:

$$E\left[\hat{\eta}_{it}(\beta) \otimes \left(1, k_{it}, l_{it}, m_{it-1}\right)\right] = 0$$

5.4.5 Markups and Markdowns à la Yeh, Macaluso and Hershbein, AER (2022)

Let Q_{it} denote the gross output of firm i at time t produced with technology:

$$Q_{it} = Q_{it}(\mathbf{X}_{it}, \omega_{it}) \tag{52}$$

where $Q_{it}(.)$ is (potentially firm-specific) technology, $\mathbf{X}_{it} = (X_{it}^1, ..., X_{it}^J)'$ is the vector of J production inputs (think in (K_{it}, L_{it}, M_{it}), $J = 3$), and ω_{it} is productivity (unobserved by the econometrician).

Let Y_{it} denote the **observed** gross output in the data, such that:

$$Y_{it} = Q_{it} \exp(\varepsilon_{it}) \tag{53}$$

where ε_{it} represents i.i.d. (zero-mean) measurement error and/or unanticipated production shocks (unobserved by the firm and the econometrician).

Let $R_{it} = P_{it}(Q_{it})Q_{it}$ denote revenues with $P_{it}(Q_{it})$ denoting the inverse demand curve assumed to be continuously differentiable and strictly decreasing ($\partial P(\cdot)/\partial Q < 0$), meaning the firm has monopolistic power over its product. Production inputs have pricing schedules $\{P_{it}^j\}_{j=1}^J$ and the adjustment cost function for input j is denoted by $\Phi_t^j(X_{it}^j, X_{it-1}^j)$.

Key assumptions: there exists at least one ***fully flexible*** input j that satisfies:

1. $\Phi_t^j(X_{it}^j, X_{it-1}^j) = 0$ (no adjustment cost)
2. $P_{it}^j(X_{it}^j) = P_{it}^j$ (no monopsony power)
3. X_{it}^j is chosen statically

A firm's monopolistic power induces a price markup over the marginal cost. If assumptions 1, 2, and 3 hold for some input j **other than labor**, say materials ($j = M$), then the markup can be computed as:

$$\mu_{it} = \frac{\theta_{it}^m}{\alpha_{it}^m} \quad \text{with} \quad \theta_{it}^m = \frac{\partial Q_{it}(\cdot)}{\partial M_{it}} \frac{M_{it}}{Q_{it}} \quad \text{and} \quad \alpha_{it}^m = \frac{P_{it}^M M_{it}}{P_{it} Q_{it}}. \tag{54}$$

A firm's monopsony power in the labor market violates assumption 2. If still assumptions 1 and 3 hold for labor ($j = L$), then the labor markdown (the ratio of MRPL over the wage)

can be computed as:

$$\nu_{it} = \frac{\theta_{it}^{\ell}}{\alpha_{it}^{\ell}} \cdot \frac{1}{\mu_{it}} \quad \text{with} \quad \theta_{it}^{\ell} = \frac{\partial Q_{it}(\cdot)}{\partial L_{it}} \frac{L_{it}}{Q_{it}} \quad \text{and} \quad \alpha_{it}^{\ell} = \frac{W_{it} L_{it}}{P_{it} Q_{it}}. \tag{55}$$

To fix ideas, consider a **Cobb-Douglas production function** with 3 inputs, (K_{it}, L_{it}, M_{it}), in an environment where **the firm has output and labor market power**. The cost minimization problem becomes:

$$\mathcal{L} = r_{it}^{K} K_{it} + W_{it}(L_{it}) L_{it} + P_{it}^{M} M_{it} + \Phi_t(K_{it}, K_{it-1}) - \lambda_{it}[K_{it}^{\theta_{it}^{k}} L_{it}^{\theta_{it}^{l}} M_{it}^{\theta_{it}^{m}} \exp(\omega_{it}) - \overline{Q}_{it}]$$

Using the classic result from profit maximization, $\lambda_{it} = C'(Q_{it}) = \frac{P_{it}}{\mu_{it}}$, the FOCs for the cost minimization problem yield the factor demand system:

$$\begin{aligned} r_{it}^{K} K_{it} + \Phi_t'(K_{it}, K_{it-1}) &= \theta_{it}^{k} \frac{P_{it} Q_{it}}{\mu_{it}} \\ \left[1 + \frac{W'(\cdot) L_{it}}{W_{it}(\cdot)}\right] W_{it}(\cdot) L_{it} &= \theta_{it}^{l} \frac{P_{it} Q_{it}}{\mu_{it}} \\ P_{it}^{M} M_{it} &= \theta_{it}^{m} \frac{P_{it} Q_{it}}{\mu_{it}} \end{aligned} \tag{56}$$

Now let $\mathbf{R}_{it}(L_{it}) \equiv rev(L_{it}; \mathbf{X}_{-L}^{*}(L_{it}))$ denote revenues when all inputs other than labor are evaluated at the optimum. The profit maximization problem is:

$$\max_{L_{it}} \mathbf{R}_{it}(L_{it}) - W_{it}(L_{it}) L_{it},$$

with first-order condition

$$\nu_{it} \equiv \frac{\mathbf{R}_{it}'(\cdot)}{W_{it}(\cdot)} = 1 + \frac{W_{it}'(\cdot) L_{it}}{W_{it}(\cdot)} \tag{57}$$

Plugging (57) in (56):

$$r_{it}^K K_{it} = \theta_{it}^k \frac{P_{it}Q_{it}}{\mu_{it}} - \Phi_t'(K_{it}, K_{it-1}) \tag{58}$$

$$\nu_{it} = \frac{\theta_{it}^l}{\alpha_{it}^l} \frac{1}{\mu_{it}} \tag{59}$$

$$\mu_{it} = \frac{\theta_{it}^m}{\alpha_{it}^m} \tag{60}$$

Key takeaways

- The cost shares method outlined in Section 5.4.3 relies on the assumptions of no adjustment costs $\Phi(\cdot) = 0$ and no monopsony power in labor ($\nu_{it} = 1$).
- This framework collapses to De Loecker et al. papers when $\nu_{it} = 1$.
- To estimate the markup μ_{it}, equation (58) has two obvious problems: first, adj. cost, second, r^K and K data issues. This is why nobody uses the FOC for capital to recover the markup.
- Equation (60) Recovers the markup μ_{it}.
- Provided the markup μ_{it}, equation (59) recovers the markdown ν_{it}.

5.4.6 Cost Shares Requires No Monopsony Power

Abstracting from capital adjustment costs ($\Phi = 0$), we have:

$$\begin{aligned} r_{it}^K K_{it} &= \theta_{it}^k \frac{P_{it}Q_{it}}{\mu_{it}} \\ \nu_{it} W_{it} L_{it} &= \theta_{it}^l \frac{P_{it}Q_{it}}{\mu_{it}} \\ P_{it}^M M_{it} &= \theta_{it}^m \frac{P_{it}Q_{it}}{\mu_{it}} \end{aligned} \tag{61}$$

Recall the idea is to recover θ_{it}^j, $j = \{k, l, m\}$ from (61) system. We can write:

$$
\begin{aligned}
\theta_{it}^{k} &= \frac{r_{it}^{K} K_{it}}{r_{it}^{K} K_{it} + \nu_{it} W_{it} L_{it} + P_{it}^{M} M_{it}} RTS_{it} \\
\theta_{it}^{l} &= \frac{\nu_{it} W_{it} L_{it}}{r_{it}^{K} K_{it} + \nu_{it} W_{it} L_{it} + P_{it}^{M} M_{it}} RTS_{it} \\
\theta_{it}^{m} &= \frac{P_{it}^{M} M_{it}}{r_{it}^{K} K_{it} + \nu_{it} W_{it} L_{it} + P_{it}^{M} M_{it}} RTS_{it}
\end{aligned}
\tag{62}
$$

So, there is a circularity problem. We cannot know the markdown ν_{it} before computing θ_{it}^{j}. But computing θ_{it}^{j} requires knowledge of the markdown.

Clearly under $\nu_{it} = 1$ (no monopsony power) we recover the ability to use cost shares, provided some assumption for the returns to scale RTS_{it}, as reported in Section 5.4.3.

5.4.7 Price Cost Margin (PCM)

Price-cost Margin is an accountability measure to assess a firm's wellness. For this indicator, we don't assume any market structure, nor impose any restriction on the firm. This is a well-used indicator since it only depends on the available information of the firm [9].

$$
\text{PCM}_{ijt} = \frac{\text{Value Added} - \text{Labor Cost}}{\text{Total Revenue}} \tag{63}
$$

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